

Escola Platform - Database Schema

Schemas	Tables	Normalized Columns	Extensions
6	17	22	uuid-oss, pgcrypto, unaccent

IDENTITY SCHEMA

Table: tenants
tenant_id (PK)
tenant_uuid (UK)
tenant_name
tenant_name_normalized
tenant_type
is_active
+ audit fields

Table: app_users
user_id (PK)
user_uuid (UK)
tenant_id (FK)
auth_user_id (UK)
full_name
full_name_normalized
email
email_normalized
user_role
is_active
+ audit fields

ASSETS SCHEMA

Table: media_files
file_id (PK)
tenant_id (FK)
storage_path

SCHOOL SCHEMA

original_name
original_name_normalized
mime_type
size_bytes
alt_text
alt_text_normalized
metadata (JSONB)
+ audit fields

Table: students
student_id (PK)
student_uuid (UK)
tenant_id (FK)
nickname + normalized
first_name + normalized
middle_name + normalized
last_name + normalized
birth_date
avatar_config (JSONB)
is_active
+ audit fields

Table: guardians
guardian_id (PK)
guardian_uuid (UK)
tenant_id (FK)
user_id (FK)
phone_number
relationship
is_primary
+ audit fields

Table: student_guardians
student_guardian_id (PK)
student_id (FK)

guardian_id (FK)
tenant_id (FK)
relationship_notes + normalized
is_authorized_pickup
+ audit fields

Table: classes
class_id (PK)
class_uuid (UK)
tenant_id (FK)
class_name + normalized
grade_level + normalized
school_year + normalized
max_students
class_config (JSONB)
is_active
+ audit fields

Table: teachers
teacher_id (PK)
teacher_uuid (UK)
tenant_id (FK)
user_id (FK)
specialization + normalized
hire_date
teacher_config (JSONB)
is_active
+ audit fields

Table: class_students
class_student_id (PK)
class_id (FK)
student_id (FK)
tenant_id (FK)
enrollment_date
status
+ audit fields

CONTENT SCHEMA

Table: modules
module_id (PK)
module_uuid (UK)
module_name + normalized
description + normalized
module_type
difficulty_level
recommended_age_min/max
display_order
thumbnail_url
module_config (JSONB)
is_published
+ audit fields

Table: activities
activity_id (PK)
activity_uuid (UK)
module_id (FK)
activity_name + normalized
description + normalized
activity_type
display_order
estimated_duration
points_reward
activity_data (JSONB)
thumbnail_url
is_published
+ audit fields

Table: assets
asset_id (PK)
asset_uuid (UK)
asset_name + normalized
asset_type
file_url
file_size_bytes

GAME SCHEMA

mime_type
duration_seconds
dimensions (JSONB)
alt_text + normalized
asset_metadata (JSONB)
is_published
+ audit fields

Table: student_progress
progress_id (PK)
progress_uuid (UK)
student_id (FK)
activity_id (FK)
tenant_id (FK)
status
score
attempts
time_spent_seconds
progress_data (JSONB)
completed_at
+ audit fields

Table: badges
badge_id (PK)
badge_uuid (UK)
badge_name + normalized
description + normalized
badge_type
icon_url
rarity
points_value
unlock_criteria (JSONB)
display_order
is_active
+ audit fields

Table: student_badges
student_badge_id (PK)
student_id (FK)
badge_id (FK)
tenant_id (FK)
earned_at
earn_metadata (JSONB)
+ audit fields

Key Relationships

From	To	Type	Description
tenants	app_users	1:N	Multi-tenant isolation
tenants	students	1:N	School students
tenants	media_files	1:N	Tenant media library
app_users	guardians	1:1	User is guardian
app_users	teachers	1:1	User is teacher
students	student_guardians	1:N	Student has guardians
guardians	student_guardians	1:N	Guardian has students
students	class_students	1:N	Student enrollment
classes	class_students	1:N	Class roster
modules	activities	1:N	Module contains activities
students	student_progress	1:N	Progress tracking
activities	student_progress	1:N	Activity completion
students	student_badges	1:N	Earned badges
badges	student_badges	1:N	Badge awards

Key Features

Feature	Description
Hybrid ID Strategy	Internal INT (student_id) + External UUID (student_uuid)
Text Normalization	22 *_normalized columns for accent-insensitive search (José → jose)
Multi-Tenancy	RLS enabled for tenant isolation via tenant_id
Audit Trail	All tables: created_at/by, updated_at/by, deleted_at (soft delete)
JSONB Columns	Flexible config/metadata storage (avatar_config, module_config, etc.)
Extensions	uuid-ossdp (UUID gen), pgcrypto (encryption), unaccent (search)